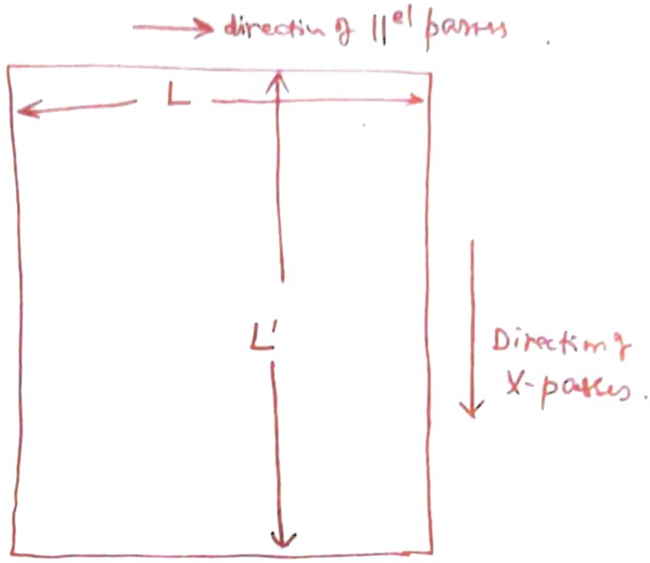
Estimation of Productivity of Rippers.

Considering the generalised case when the ripping is done in $||^{el}$ and X-passes.



given parameters

V_r = ripping velocity in top gear. m/s.

L = length of $||^{el}$ passes.

L' = Length of cross-passes.

T = time of ripper to travel back to next furrow. (seconds) during $||^{el}$ pass

~~Assumption → L & L' are known that L' = (1.2 to 1.5) L~~

T' = time of ripper to travel back to next furrow in. (Second) during X-pass.

C = ? Spacing between two passes in. $||^{el}$

C' = ? Spacing between two passes in X-direction.

Now:

No of passes required in $||^{el}$ cut $= \left(\frac{L'}{C}\right)$.

No of passes. required in X-cut $= \left(\frac{L}{C'}\right)$.

$$\text{Total time in } ||^{el} \text{ pass} = \left[\frac{L'}{C} \times \frac{L}{V_r} + T \cdot \frac{L'}{C}\right]$$

$$\text{Total time in X-pass} = \left[\frac{L}{C'} \times \frac{L'}{V_r} + T' \frac{L}{C'}\right]$$

$$\therefore \text{Total time for both } ||^{el} \text{ \& X-pass} = \left[\frac{LL'}{CV_r} + \frac{TL'}{C} + \frac{LL'}{C'V_r} + \frac{T'L}{C'}\right]$$

$$= \left[\frac{LL'}{V_r}\left(\frac{1}{C} + \frac{1}{C'}\right) + \frac{TL'}{C} + \frac{T'L}{C'}\right]$$

Now total volume excavated in $||^{el}$ & X-pass

$$= h_t \times L \times L' \quad m^3 .$$

$\therefore$ Theoretical productivity $\left(\frac{m^3}{s}\right) = \dfrac{h_t \times LL'}{\left[\frac{LL'}{v_r}\left(\frac{1}{c}+\frac{1}{c'}\right) + T\left(\frac{L'}{c}\right) + \frac{T'L}{c'}\right]}$

If L' is chosen such that $(T = T')$

Then.

Theoretic produc. $\left(\frac{m^3}{s}\right) = \dfrac{h_t \cdot LL'}{\frac{LL'}{v_r}\left(\frac{1}{c}+\frac{1}{c'}\right) + \left(\frac{TL'}{c} + \frac{TL}{c'}\right)}$

$$= \frac{h_t}{\frac{1}{v_r}\left(\frac{1}{c}+\frac{1}{c'}\right) + T\left(\frac{1}{Lc} + \frac{1}{L'c'}\right)}$$

So

practical hourly productivity considering K_u (ripper utilization factor) (0.7 to 0.8)

$$Q_{r\,(||^{el}+X)} = \frac{3600 \times K_u \times h_t}{\frac{1}{v_r}\left(\frac{1}{c}+\frac{1}{c'}\right) + T\left(\frac{1}{cL} + \frac{1}{c'L'}\right)} \quad m^3/hr.$$

Productivity Calculation for ||el pass. only.

No of passes in ||el $= \frac{L'}{c}$.

Volume loosened in ||el passes $= \left(h_{eff} \times L \times L'\right)$

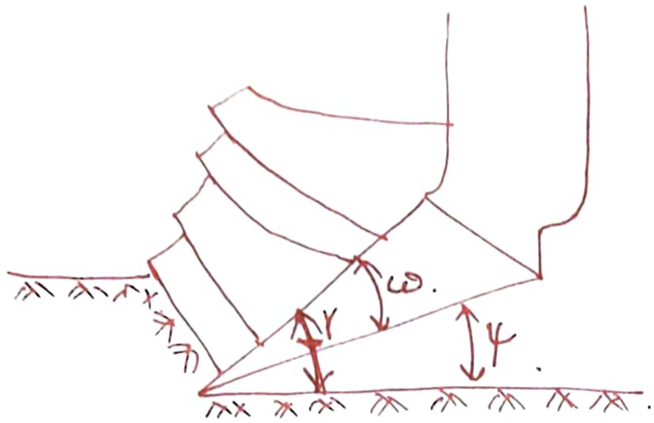
Total time taken in ||el passes $= \left(\frac{L'}{c}\right)\frac{L}{V_r} + \frac{TL'}{c}$

$$Q\left(\frac{m^3}{s}\right) = \frac{h_{eff} \times LL'}{\frac{LL'}{cV_r} + \frac{TL'}{c}}.$$

$$= \frac{h_{eff}}{\frac{1}{cV_r} + \frac{T}{Lc}} = \frac{h_{eff} \cdot \times c}{\left(\frac{1}{V_r} + \frac{T}{L}\right)}.$$

∴ Hourly productivity considering effective ripper utilization (0.7-0.

$$Q_{r\,(||el)} = \left(\frac{3600 \times c \times h_{eff} \times K_u}{\frac{1}{V_r} + \frac{T}{L}}\right)$$



$\omega = 20$ to 30°.

γ = cutting angle.

ω = tip angle

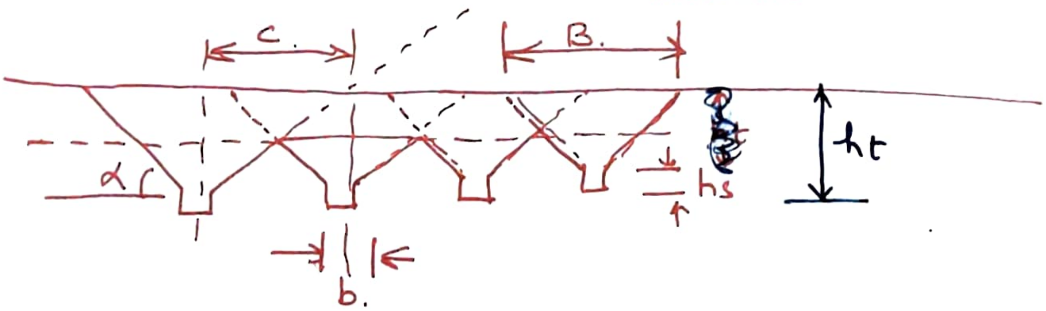
ψ = tooth clearance angle.

$\psi = 5$ to 7° (igneous/metamorphic roc

$\psi = 8$ to 10° = frozen rock mass.

γ is automatically desired from ($\psi + \omega$)

smaller clearance angle causes crushing of rock mass.



Slot width = b. (close to thickness of tooth tip (b_t)).

h_s = slot height. (0.15 to 0.2 times ripper tooth penetration h_t)

α = slope of sidewall of the cut (40-60°) depends on rock mass characteristic & tooth tip geometry

rock mass loosening ability of a ripper is specified by

- possible penetration (h_t) of ripper tooth into the loosened rock-mass

monolithic rock mass → loosening due to overcoming tensile strength

jointed rock mass → overcoming cohesion bet str. blocks.

C = distance between adjacent passes.

- Set with an aim to ensure required lump size.
- Sufficient loosening depth.

(1)

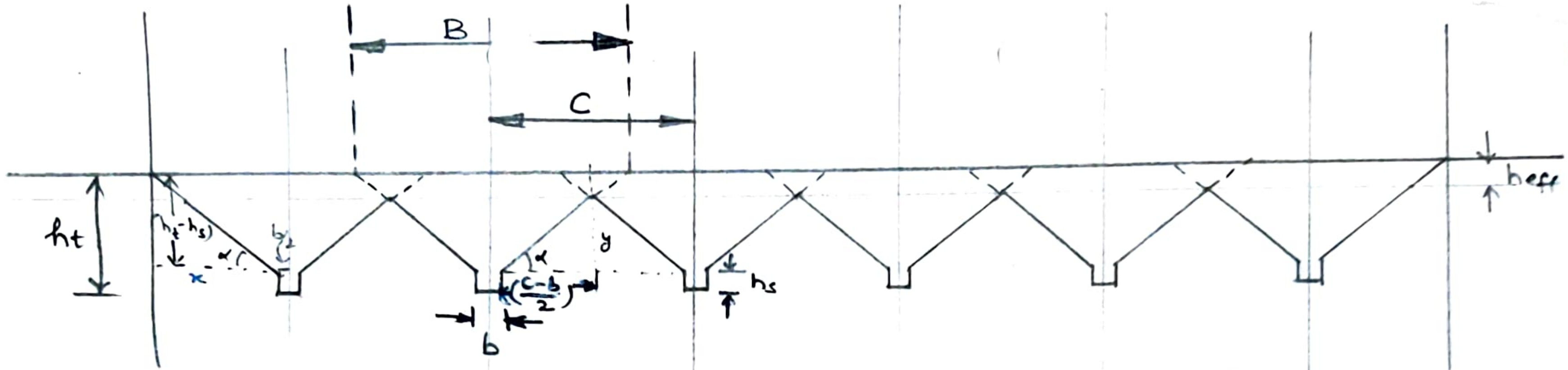


Figure :- for $\parallel^{el}$ cuts made by ripper.

Rippers

$$\frac{h_t-h_s}{x} = \tan\alpha \qquad \frac{y}{\left(\frac{C-b}{2}\right)} = \tan\alpha \qquad \frac{y}{\left(\frac{C-b}{2}\right)} = \tan\alpha .$$

$$\Rightarrow x = (h_t-h_s)\cot\alpha . \qquad y = \left(\frac{C-b}{2}\right)\tan\alpha .$$

Not. used directly

$$\frac{B}{2} = (h_t-h_s)\cot\alpha + \frac{b}{2}$$

$$\Rightarrow \boxed{B = 2(h_t-h_s)\cot\alpha + b}$$

B = Width of furrow.

C = distance between adjacent passes.

Considering n-parallel passes, The total volume of material.

$$V = [(n-1)C + B]\, h_{eff} \qquad \text{—— (1)}$$

$$h_{eff} = h_t - h_s - y$$

$$= (h_t-h_s) - \left(\frac{C-b}{2}\right)\tan\alpha . \qquad \text{—— (2)}$$

Using Equation ① & ② we get.

$$v = \left[(n-1)C + B\right]\left[(h_t - h_s) - \left(\frac{C-b}{2}\right)\tan\alpha\right]$$

Now expanding we get.

$$V = (n-1)(h_t - h_s)C - \frac{(n-1)C(C-b)}{2}\tan\alpha$$
$$+ B(h_t - h_s) - B\times\left(\frac{C-b}{2}\right)\tan\alpha$$

or: $$V = (n-1)(h_t - h_s)C - \frac{(n-1)C^2}{2}\tan\alpha + \frac{(n-1)bc}{2}\tan\alpha$$
$$+ B(h_t - h_s) - \frac{Bc\tan\alpha}{2} + \frac{bB}{2}\tan\alpha.$$

differentiating.

$$\frac{dV}{dc} = (n-1)(h_t - h_s) - (n-1)C\tan\alpha + \frac{(n-1)}{2}b\tan\alpha - \frac{B\tan\alpha}{2}$$
$$= 0$$

$$\Rightarrow (n-1)(h_t - h_s) - (n-1)C_{opt}\tan\alpha + \frac{(n-1)b\tan\alpha}{2} - \frac{B\tan\alpha}{2} = 0$$

or, $$(n-1)(h_t - h_s) + \frac{(n-1)}{2}b\tan\alpha - \frac{B\tan\alpha}{2} = (n-1)C_{opt}\tan\alpha$$

or, $$C_{opt}\times[n-1]\tan\alpha = (n-1)(h_t - h_s) + \frac{(n-1)b}{2}\tan\alpha - \frac{B\tan\alpha}{2}$$

$$\Rightarrow C_{opt} = (h_t - h_s)\cot\alpha + \frac{b}{2} - \frac{B}{2}\times\frac{1}{(n-1)}$$

Now as $n \to \infty \Rightarrow \frac{B}{2}\times\frac{1}{n-1} \to 0$.

$$\therefore C_{opt} = (h_t - h_s)\cot\alpha + \frac{b}{2}$$

- Now using this value of C_{opt} in Equation (2) we get.

$$h_{eff} = (h_t - h_s) - \frac{(h_t - h_s)\cot\alpha + \frac{b}{2} - b}{2}\tan\alpha .$$

$$= (h_t - h_s) - \frac{(h_t - h_s)\cot\alpha \cancel{+} \frac{b}{2}}{2}\tan\alpha$$

$$= (h_t - h_s) - \frac{(h_t - h_s) \cancel{+} \frac{b}{2}\tan\alpha}{2}$$

$$= \frac{h_t - h_s}{2} + \frac{1}{2}\times\frac{b}{2}\tan\alpha .$$

$$\therefore \quad h_{eff} = \frac{(h_t - h_s)}{2} + \frac{b}{4}\tan\alpha .$$

(2 1/2) marks for explaing the understandig of application.

The Concept of minimum tipping load & max operative load is used to find out the suitable bucket size of a particular front end loader for a given job condition.

Question (2) Estimate the HEMM required in a surface iron-ore mine for producing 10 million ton of ore per year. The average stripping ratio is 0.8 ton/ton. The mine will be worked by a shovel dumper combination with a bench height of 12 m. (Assume ~~of~~ relevant data wherever necessary and mention them)

Solution -(2)

Given Condition

(i) Production of ore per year = 10×10^6 ton

(ii) Average stripping ratio = 0.8 ton/ton

(iii) Bench height = 12 m (1/2) marks for write the given condition.

Assumption

Mining parameter (1/2) marks for realistic assumptions.

(i) Bank density of ore = 3.6 ton/m3

(ii) Bank density of waste = 2.8 ton/m3

2½ marks for understanding explaining the application.

Shovel parameters

(i) Bucket capacity = $5 m^3$

(ii) Bucket fill factor = 0.8

(iii) Swell factor = 25%.

2 marks for realistic assumption

(iv) Standard cycle time of shovel. } = 30 s. (90° swing)

(v) Overall efficiency = 0.56

(vi) Working Hrs = 300×3×8 = 7200 Hrs

Dumper parameter :

(i) rated capacity of dumper } = 50 ton.

(ii) Spotting time of dumper = 0.3 min.

2 marks for realistic assumption.

(iii) Dumping time of dumper = 1.5 min

(iv) Load travel speed of dumper } = 20 km/hr.

(v) Empty travel speed of dumper = 30 km/hr.

(vi) Haul distance (one way) both ore + waste. } = 1 km.

Calculation

Production of ore per year = 10×10^6 ton.
Handling of ~~w~~ waste per yr = $0.8 \times 10 \times 10^6$

$= 8 \times 10^6$ ton.

Total material to be handled per year } = 18×10^6 ton.

average density of composite (ore & waste) } $= \dfrac{(1+SR) \times \rho_o \times \rho_w}{(\rho_w + SR \cdot \rho_o)}$ ton/m3.

3 marks for evaluating the average density.

$$= \frac{(1+0.8)(3.6)(2.8)}{(2.8 + 0.3 \times 3.6)}$$

$= 3.279$ t/m3.

Estimation the No of shovels

Required material handling per year (te) = 18×10^6
Average density of ore & composite = 3.279

4 marks for calculating No. of shovel.

Effective producing per shovel per year (te) } $= B_c \times B_f \times S_f \times \dfrac{3600}{t_c \times F_s} \times 7200 \times \eta \times \rho_{av}$

$$= 5 \times 0.8 \times \left(\frac{100}{100+25}\right) \times \frac{3600}{30 \times 1} \times 7200 \times 0.56 \times 3.279$$

$= 5076836.35$ te.

Calculation

No of shovels Require = $\frac{18 \times 10^6}{5076836.35}$

= 3.546

$\boxed{\text{Actual no of Shovels Reqd} = 4}$

Calculating No of Dumpers.

Effective Cap. of dumper (te) = 50

ton per pass of Shovel = $B_c \times B_f \times S_f \times \rho_{av.}$

$= 50 \times 0.8 \times \left(\frac{100}{100+25}\right) \times 3.279$

$= 10.493$

No of passes Reqd to fill the dumper } = $\frac{50}{10.493}$

= 4.77

Actual No of pass reqd to fill the dumper } = 4.0

2 marks for calculating No. of passes.

Total loading time of dumper (min) = 2
Spotting time (min) = 0.3
Dumping time (min) = 1.5
Load travel time = $\frac{60 \times 1}{20}$ = 3.0
Empty travel time = $\frac{60 \times 1}{30}$ = 2.0

Total cycle time of dumper (min) = 8.8.

No of Dumpers reqrd = $\frac{\text{Total time}}{(\text{Spotting time + loading time})}$

4 marks for Calculating No of dumps. = $\frac{8.8}{(2+0.3)}$ = 3.83

~~& Summarising.~~

$\boxed{\begin{array}{l}\text{Actual No of dumps per shovel} = 4 \\ \text{So Total No of dumpers Reqrd} = 4 \times 4 = 16\end{array}}$

Summary of HEMM

Item	Capacity	Qty	Total	Remarks	Spare
Shovel	5 m3	4	6		2
Dumper	50 te	16	24		8
Drill	250mm	4	6		2
Dozer	250/350HP	8	12		4
FEL	3.2m3	2	3		1
Motor grader		1	2		1
Crane		1	2		1
Water sprinkler	5KL	1	2		1

2 marks for Summarising the table.

if unrealistic wrong assumption is used in calculations but steps are ok. each step will be evaluated at 50% marks.

Q. 3. (a) What is a "box-cut"? What are its objectives? With the aid of suitable sketches, describe the different types of box-cut and also indicate their respective applicability and limitations.

(b) Discuss briefly the factors that influence the choice of location of box-cut in case of sub-surface deposits.

Ans:- Box-cut is the initial/first cut given for the physical development of a mine. The name "box-cut" has been attributed to this cut as this cut generally looks like a top open box having an inclined floor & walls on three sides (front wall & two side walls) in case of opening up of sub-surface deposits.

2 marks for defn & expln

Objectives of a box cut

- to reach the ore body/coal seam
- to provide smooth entry to the pit.
- to provide space for development of workings (production benches.

2 marks for enumerating objective.

Types and applicability of Box-cut

(1) Internal box cut:

When the box cut is located fully or partially on mineralised zone, it is called an internal box-cut.

(2)

Sl. No. 16-17/ 050128

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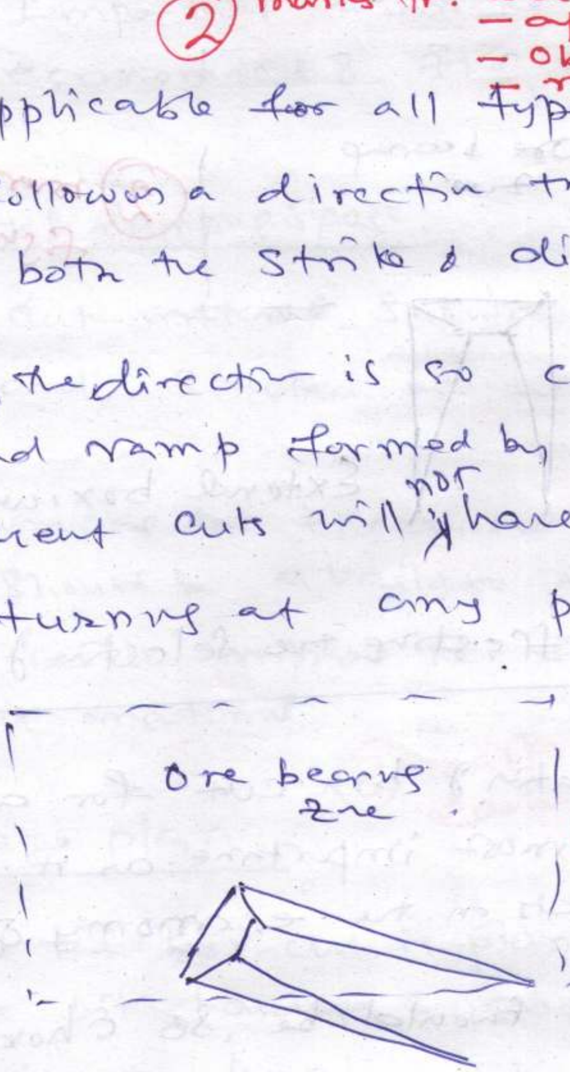
(2) marks fr. - defn - applicibily. - obliquity - reasons.

This is applicable ~~for~~ all types of deposit.

The cut follows a direction that is usually oblique to both the strike & dip direction.

General all, the direction is so chosen that the haul road ramp formed by this cut and subsequent cuts will not have unnecessary steep turning at any position.



(2) marks for figure.

Internal Box cut.

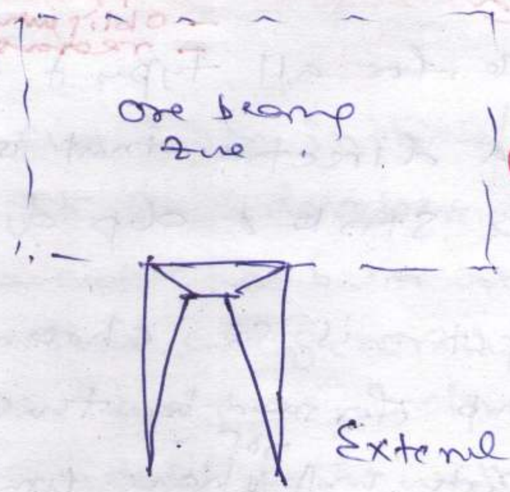
External Box-cut

When the box cut is placed totally outside the mineralized zone, it is called an external box-cut.

(2) marks for defn explen & applicabily.

This is applicable only for shallow and gently dipping bedded deposits.

This cut is generally located at the middle of the rise-most side.



Extenal box cut.

2 marks fr. figure.

Factors Affecting the selection of Box-cut.

Selection of location of box-cut for any surface mine is of utmost importance as it has far reaching effects on the economy of the mine.

The location should be so chosen that there is a min of foreseeable trouble with uncovering of the ore body and a max. of economy.

Factors

(1) Site accessibility; easily accessible/reachable for men & machinery.

(2) Minimum Excavation Reqmt.

- material to be excavated to reach the ore body will be as less as possible.
- this ensures early production of ore from the mine.
- Important from the view point of the economics of the mine operation.

(3) Availability of dumping space:

- The cut material should preferably be handled only once and stacked off the mineral bearing zone.
- for dumping the excavated material, sufficient space should be available nearby
- this will improve the economy of transport of cut material.

(8) marks for each point.

(4) Pit haulage plan:

- if the box cut is planned to be a part of the pit haulage system, it must match with the haulage plan for the complete pit.
- the roads as far as practicable, should be put in but once and used as much as possible.

(5) Overall mine plan:

- The box cut location should be so chosen that it matches to in the overall mine plan including the waste dump locations.
  - Sites for infrastructural facilities.
  - Sites for other facilities.
- it should not hinder the pit progress/expansion at a later date.
- the location should be such that the overall economy of ore and waste transport over the whole mine life is maintained.

(6) Water Condition.

- Surface drainage must be diverted, preferably by gravity.
- if possible, the location of the cut should be so chosen that the problems of accumulation of water from surface water and/or ground water are not there or minimum.

(7) Geological Disturbance

- free of any geological discontinuity and structural disturbances.

(8) Reclamation Requirement:

- The reclamation requirements may dictate that the box-cut material should be dumped / emplaced at particular location.

Ques-(4) (a) With the help of suitable sketches, describe the lateral block cut method operation of bucket wheel excavator.

(b) Calculate the theoretical hourly output for a bucket wheel excavator from the following given data.

(i) No of buckets on wheel = 10.

(ii) Individual bucket cap = 1200 lts.

(iii) Wheel diameter = 9 m

(iv) Cutting speed at the knife edge } = 2.1 m/s.

(V) % Swelling of the material to be excavated } = 25

(6) Estimate the no. of drills required to achieve a production of 5 million ton/year of coal from a surface mine having coal seam of thickness 10m. The coal seam is flat and horizontal and the stripping ratio of the mine is 3 m3/ton. (Assume relevant data wherever and mention them).

Ans:-

(2) marks for Calculating the OB thickness

$$\boxed{\text{OB thickness} = SR \times (\text{Thick of Coal seam}) \times \rho_{coal}}$$

OB thm = 3 × 10 × 1.6 = 48 m.
∴ 4 Bench of 12 m each.

Given Condition:

(1) marks for mentioning the given condition & assumption of realistic density

1. Coal production per year = 5×10^6 te.
2. SR = 3 m3/ton
3. Coal seam thickness = 10 m.
4. Insitu density of Coal = 1.6 ton/m3.

Assumptions.

	Coal Seam	OB Bench
1. Drill dia.	150	250
2. Burden.	4.5	6
3. Spacing.	6.5	8
4. Bench height.	10	12 m
5. Length of drill rod	6	7

		Coal	OB
6.	Time to set drill m/c (min)	6	7
7.	Time to set drill rod (min)	3	3
8.	Rate of drilling (min/m)	2	4
9.	Subgrade drill	0	10 %
10.	Overall utilization	0.6	0.6
11.	Drilling hrs per (yr)	3000	3000

2½ + 2½

for realistic assumptions for OB & coal

Calculation:-

Estimation of Drilling requirement/year.

Volume of Coal handling/year = 3125000

Vol of coal handled/hole (m^3) = 292.5

No of holes Reqd/year = 10683.76

Actual No of holes Reqd/yr = 10684

No of feed Reqd = 1.66 ≈ 2

time to drill one hole = 29 min.

3 marks for calculating time to drill

Capacity of drill m/c in coal.

Actual holes / hr = 2.0690.

Actual Hole per yr = 3724.14.

No of drills = 2.86 3 marks for Calculating no of drills.

Actual No of drills = 3.

Estimating OB. Drills

a) Drilling requirement per year.

Vol of OB handled / yr (Prodn x SR) = 1,50,00,000.

Vol of OB handled / ~~hole~~ hole = 576

No of holes Reqd = 26041.667.

Actual No of holes Reqd = 26042.

Depth of drilling in OB = 13.2 m.

No of feed Reqd = 1.88 ≃ 2.

No of feed change Reqd = 1

3 marks for calculating time to drill

time to drill one hole = 62.8 mne

Cap of drill mle in org : if steps are correct & using data from upstream is used it will be evaluated at 50%.

Actual holes / hr = 0.955

Actual hours / yr = 1719.75

No of drill reqd = 15.14 (3) marks for calculating the No of drills including summary.

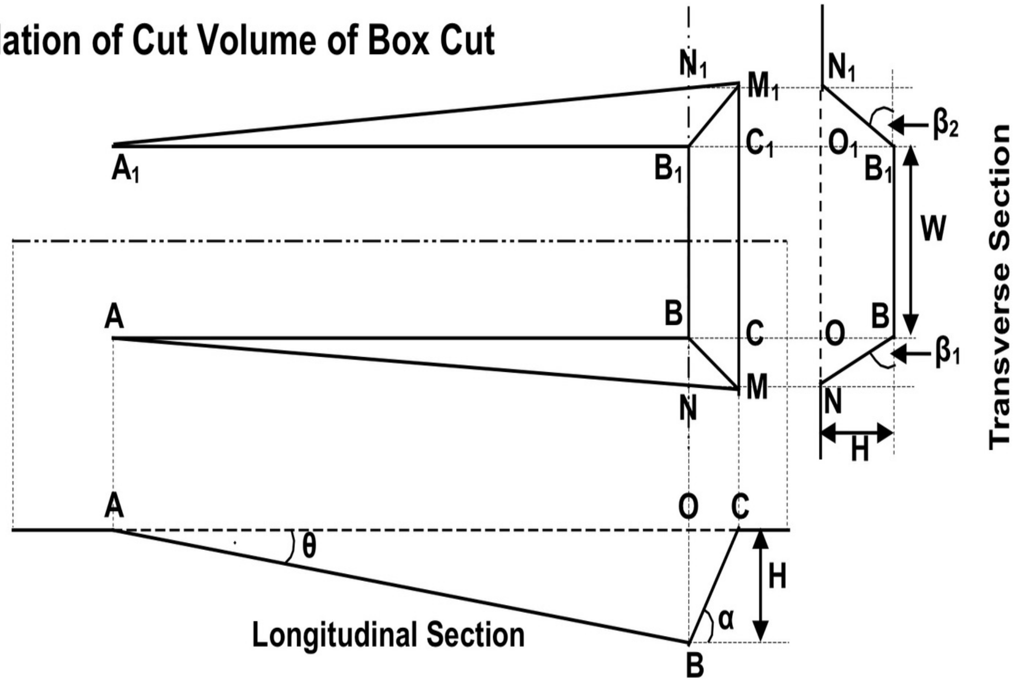
Actual no of drills = 16

∴ OB - Drills = 16
Coal Drills = 3

Surface Mining

* Calculation of Cut Volume of a box cut :-

2.1.3. Calculation of Cut Volume of Box Cut



• The points O & O_1 are the projections of B & B_1 on the surface.

→ Purpose is to find AC :-

Consider ΔAOB; $\tan\theta = \frac{OB}{AO}$

$$\Rightarrow AO = \frac{OB}{\tan\theta} = \frac{H}{\tan\theta} \quad \text{— (i)}$$

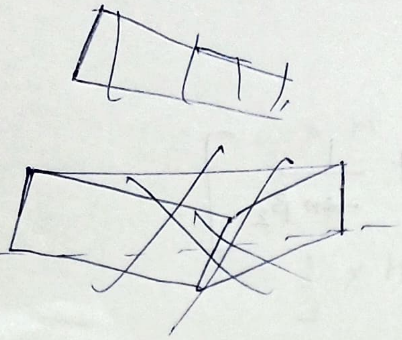
for ΔBOC; $\tan\alpha = \frac{OB}{OC} = \frac{H}{OC}$ — (1)

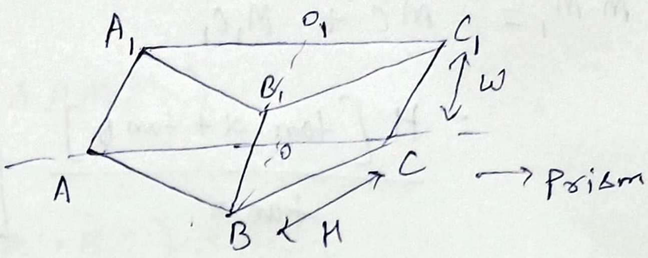
$$\Rightarrow OC = \frac{H}{\tan\alpha} \quad \text{— (ii)}$$

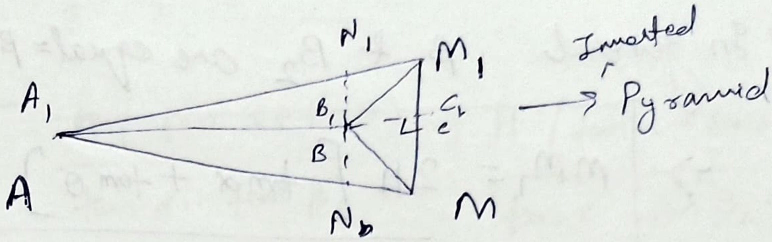
$\because$ $AC = AO + OC$

$$AC = H\left[\frac{1}{\tan\theta} + \frac{1}{\tan\alpha}\right]$$

$$= \frac{H\left[\tan\alpha + \tan\theta\right]}{\tan\alpha \cdot \tan\theta} \quad \text{— (2)}$$

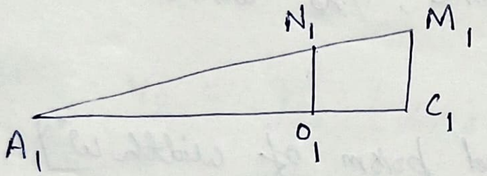


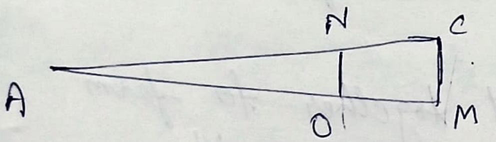




• Consider the surface projection (Plan Surface) ~~ABC~~

ADCMNA & $A_1O_1C_1M_1N_1A_1$





from transverse section

ΔOBN :-

$\tan\beta_1 = \frac{OB}{ON}$

$\Rightarrow ON = \frac{H}{\tan\beta_1}$ — (3)

$\Delta O_1B_1N_1$:-

$\tan\beta_2 = \frac{O_1B_1}{O_1N_1}$

$\Rightarrow O_1N_1 = \frac{H}{\tan\beta_2}$ — (4)

From $\sim\Delta^s$ AON & ACM :

$$\frac{MC}{ON} = \frac{AC}{AO}$$

$$\Rightarrow MC = \frac{AO+OC}{AO} = 1 + \frac{OC}{AO}$$

$$\Rightarrow MC = ON\left[1 + \frac{OC}{AO}\right] = \frac{H}{\tan\beta_1}\left[1 + \frac{H}{\tan\alpha} \times \frac{\tan\theta}{H}\right]$$

• $$MC = \frac{H[\tan\alpha + \tan\theta]}{\tan\alpha \cdot \tan\beta_1}$$

Similarly; M_1C_1 can be defined by :-

• $$M_1C_1 = \frac{H[\tan\alpha + \tan\theta]}{\tan\alpha \cdot \tan\beta_2}$$

- $MM_1 = MC + M_1C_1$

$$= \frac{H\left[\tan\alpha + \tan\theta\right]}{\tan\alpha}\left[\frac{1}{\tan\beta_1} + \frac{1}{\tan\beta_2}\right]$$

→ In general, β_1 & B_2 are equal $= \beta$.

$$\Rightarrow \boxed{MM_1 = \frac{2H\left[\tan\alpha + \tan\theta\right]}{\tan\alpha \tan\beta}}$$

★ For calculation of the cut volume, the whole volume is divided into 3-segments :-

1. segment $ABCC_1B_1A_1$ [Inverted prism of width w]
2. segment $ABCMN$
3. segment $A_1B_1C_1M_1N_1$

$\Big\}$ → Joint together to form an inverted pyramid with Δ^r base AMM_1A_1.

- Volume of Prism $(ABCC_1B_1A_1) = V_1$

$V_1 = w \times Ar(\Delta ABC)$

$= w \times \frac{1}{2} \times (AC \times OB)$

$= \frac{w}{2}\left[\frac{H(\tan\alpha + \tan\theta)}{\tan\alpha \cdot \tan\theta} \times H\right]$

$$\boxed{V_1 = \frac{H^2 w}{2} \frac{(\tan\alpha + \tan\theta)}{\tan\alpha \cdot \tan\theta}}$$

• Volume of inverted pyramid = V_2

$$V_2 = \frac{1}{3} \times H \times \text{Area}(\Delta AMM_1)$$

$$= \frac{1}{3} \times H \times \frac{1}{2} \times (MC + M_1C_1) \times AC$$

$$= \frac{H}{6}\left[\frac{H(\tan\alpha + \tan\theta)}{\tan\alpha \cdot \tan\beta_1} + \frac{H(\tan\alpha + \tan\theta)}{\tan\alpha \cdot \tan\beta_2}\right]$$

$$\times \frac{H(\tan\alpha + \tan\theta)}{\tan\alpha \cdot \tan\theta}$$

$$V_2 = \frac{H^3}{6}(\tan\alpha + \tan\theta)^2\left[\frac{1}{\tan\beta_1} + \frac{1}{\tan\beta_2}\right] \times \frac{1}{\tan^2\alpha \cdot \tan\theta}$$

when $\beta_1 = \beta_2 = \beta$ then

$$\boxed{V_2 = \frac{H^3}{3}\frac{(\tan\alpha + \tan\theta)^2}{\tan^2\alpha \cdot \tan\theta}}$$

→ Total Cut Volume = $V_1 + V_2$

$$\boxed{V = \frac{H^2 w}{2}\left[\frac{\tan\alpha + \tan\theta}{\tan\alpha \tan\theta}\right] + \frac{H^3}{3}\frac{(\tan\alpha + \tan\theta)^2}{\tan^2\alpha \cdot \tan\theta \cdot \tan\beta}}$$

or,

$$\boxed{V = \frac{H^2(\tan\alpha + \tan\theta)}{\tan\alpha \cdot \tan\theta}\left[\frac{w}{2} + \frac{H}{3}\frac{(\tan\alpha + \tan\theta)}{\tan\alpha \cdot \tan\beta}\right]}$$

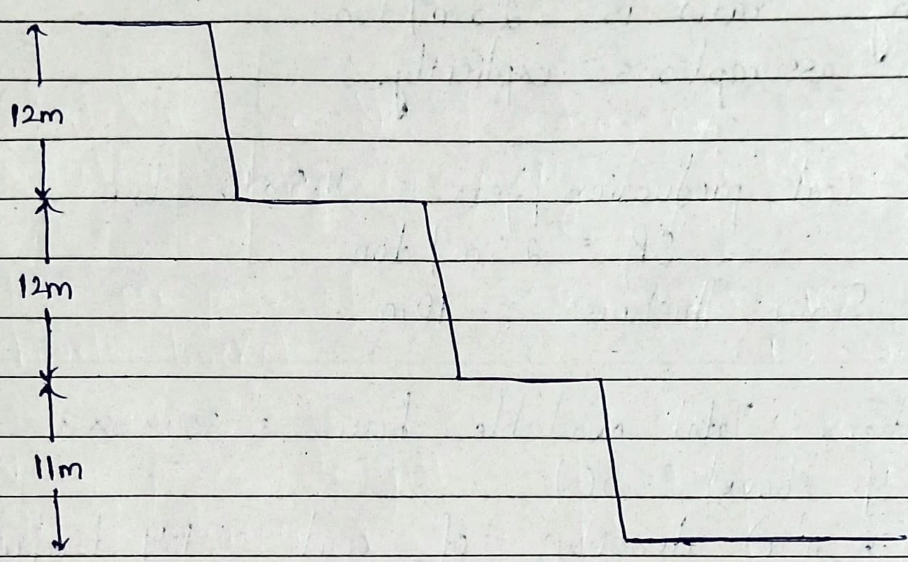
Estimate the HEMM required in a surface mine producing 4.5 million ton of coal and worked by shovel dumper combination. The average seam thickness is 10m and the average stripping ratio is $2.5 m^3/ton$.
Mention assumptions explicitly.

Given : Coal production/year = 4.5×10^6 ton
SR = $2.5 m^3/ton$
Seam Thickness = 10 m

Assumptions : Total available hours : $300 \times 3 \times 8 = 7200$
OEE of shovel = 56%
Bank (in-situ) density of coal = $1.4 \ ton/m^3$
Bucket capacity of coal : $5 m^2$
" " " OB = $5 m^2$
Bucket fill factor = 0.85 (coal)
" " " = 0.80 (OB)
% swell for OB & Coal : 20%
Back density of OB = $2.6 \ t/m^3$
Angle of swing for coal & OB = 90°
Standard time for both the shovels = 30 s
Rated capacity of dumper (both Coal & OB) = 50 ton
Spotting time (OB & coal) = 0.3 min
Dumping time (OB & coal) = 1.5 min
Load travel speed of dumpers (coal & OB) = 20 Km/hr
Empty " " " " = 30 Km/hr
Haul distance (one way for both coal & OB) = 1 Km

Average OB Thickness = Seam thickness × S.R. × S coal
= 10 × 2.5 × 1.4 = 35 m

This may be handled in 3 benches each of height 12m, 12m & 11m.



Calculations for No. of shovels

Required Production / year = 4.5×10^6 tons

Efficient production / Shovel / year $= \dfrac{0.56 \times 5 \times (100/100+20) \times 0.85 \times 3600 \times 14 \times 7200}{30}$

= 2399040 ton

No. of working shovels required $= \dfrac{4.5 \times 10^6}{2399040} \approx 1.82$

Actual No. of shovels (coal) required = 2

For OB shovels

Excavation requirement = $4.5 \times 10^6 \times 2.5 m^3$

DATE
PAGE NO.

$$\frac{5 \times 0.8 \times \left(\frac{100}{100+20}\right) \times 3600 \times 7200 \times 0.56}{30}$$

$1612800 \; m^3$

No. of shovel required $= \frac{4.5 \times 10^6 \times 2.5}{1612800} = 6.97$

Actual no. of shovel $= 7$

No. of dumpers for coal

Effective dumper Capacity $= 50 \times \frac{1.4}{2.6} = 2.7$ ton

$\frac{\text{ton}}{\text{pass of shovel bucket}} = 5 \times 0.85 \times 1.4 \times \frac{100}{100+20} = 4.958$

No. of passes required to load dumper $= \frac{27}{4.96} = 5.44$

Actual no. of passes $= 5$

Dumper loading time $= 5 \times 30\,s = 2.5$ min

Spotting time $= 0.3$ min

Load travel time $= 1/20 \times 60 = 3$ min

Dumping time $= 1.5$ min

Empty travel time $= 1/30 \times 60 = 2$ min

Total no. of dumper $= \frac{\text{total cycle time}}{\text{spotting time}} + \text{loading time}$

$= \frac{9.3}{2.8} = 3.22$

DATE
PAGE NO.

Actual No. of dumpers $= 4$

Waiting time for each dumper $= 4 \times 2.8 - 9.3$
$= 1.9$ min

Total no. of dumper required for coal $= 4 \times 2.8$
$= 11.2$ n

Calculation for No. of dumpers for Overburden

Effective Dumper capacity $= 50$ ton

Tons per pass of shovel bucket $= 5 \times 0.8 \times 2.5 \times \frac{100}{100+20} = 8.666$ tons

No. of passes to fill a dumper $= \left[\frac{50 \times 2.5}{2.6}\right] \times \frac{1}{8.666} = 5.4545$

ROUND UP
ROUND DOWN

Actual No. of passes to fill Dumper $= 5$

Dumper loading time $= 5 \times 30s = 2.5$ min
Spotting time $= 3$ min
Dumping time $= 1.5$ min
Empty travel time $= 2$ min

Total No. of Dumpers required per shovel in overburden $= \frac{0.3 + 2.5 + 3 + 1.5 + 2}{2.5 + 0.3}$

$= \frac{9.3}{2.8} = 3.32$

Total no. of required shovel $= 4$

Waiting time of each dumper $= (4 \times 2.8 - 9.1)$
$= 1.9$ min

Total no. of dumpers required for OB $= 7 \times 4 = 28$

Waiting time for each dumper = $(4 \times 2.8 - 9.3)$ min

= 1.9 min.

∴ Total No. of Dumpers Required for OB = 7×4

= 28

* Thumb Rule for HEMM fleet Estimation:-

1. No. of drills = No. of Shovels.
2. No. of Dozers = 0.5 per working shovel + 1 to 1.5 for Sunday Jobs
+ 1 or 2 at Coal Stock yard
+ 1 or 2 at waste Dump.
3. No. of working FEL (Front End Loader) = 1 or 2 supporting at pit
1 to 3 based on production level at Stockyard.
4. No. of Motor grader = 1 or 2 for Road Maintainance
5. No. of Cranes Required = 1 or 2 depending on production requirement
6. No. of water sprinkles = 1 or more depending on production requirement.

- Light Vehicles,

a) Diesel Bowger

b) ~~Sup~~ Manager's Jeep

c) ~~Supervisors~~ Supervisor's Jeep.

d) Small truck / Canter / Campar

* Statement of HEMM :-

Sl. no	Item	Capacity	Working qty	Spare	Total	Remark / Spare
1.	Coal Shovel	5 m^3	2	1	3	No. of SLD
2.	Dumper (Coal Shovel)	50 ton	8	4	12	
3.	Drill for coal face	150 mm	2	1	3	
4.	OB Shovel	5 m^3	7	2	9	
5.	Dumper for OB	50 ton	28	14	42	
6.	Drill for OB	250 mm	7	2	9	
7.	Dozer	250/350 HP	9	2	11	
8.	FEL	6 m^3	3	1	4	
9.	Motor grader		2	1	3	
10.	Crane		2	1	3	
11.	Water sprinklers		2	1	3	

Q. Estimate the HEMM required in a surface iron ore mine for producing 6.0 million ton of ore per year. The average stripping ratio is 0.3 ton / ton. The mine will be worked with shovel Dumper combination with a bench height of 12 m.

(ton / ton → metal mine
(m^3 / ton → coal mine)

Soln :- Given Condition :-

(i) Production of ore per year = 6×10^6 ton
(ii) Avg SR = 0.3 ton / ton
(iii) Bench ht = 12 m

Assumptions :-

Mining parameters :

(i) Bank Density of ore = 3.6 ton / m^3
(ii) Bank Density of waste = 2.8 ton / m^3

Shovel Parameters :-

(i) Bucket capacity = 5 m^3
(ii) Bucket fill factor = 0.80
(iii) Swell factor = 25 %
(iv) Std. cycle time of shovel = 30 s
(v) OEE = 0.56
(vi) Working Hrs = 300 × 3 × 8
= 7200 hrs

Dumper parameters :-

(i) Rated capacity of Dumper = 50 te
(ii) Spotting time of dumper = 0.3 min
(iii) Dumping time for Dumper = 1.5 min
(iv) Load travel speed of dumper = 20 km / hr
(v) Empty travel speed of dumper = 30 km / hr
(vi) Haul Distance (one way) for both wasted ore = 1 km.

Density :- ρ_w ρ_{ore}

p = production (ton) $\frac{m}{V} = \rho$

Mass = $\underbrace{P}_{ore} + \underbrace{P \cdot SR}_{waste}$

Volume = $\frac{P}{\rho_o} + \frac{P \cdot SR}{\rho_w}$

Avg. Density = $\frac{Mass}{Volume} = \frac{P + P \cdot SR}{\frac{P}{\rho_o} + \frac{P \cdot SR}{\rho_w}}$

$\Rightarrow \boxed{\rho_{avg} = \frac{(1 + SR)\,\rho_o \rho_w}{\rho_w + \rho_o SR}}$

Average density & composite ore waste.

• Calculations :-

Production of per year = 6×10^6 tonn

Handling of waste per year = $0.3 \times 6 \times 10^6$ ton

= 1.8×10^6 ton

Total material to be handled per yr = 7.8×10^6 ton

= $1.8 + 6$ =

Avg. density of composite ore & waste $\Big\}$ = $\dfrac{(1+0.3) \times 3.6 \times 2.8}{(2.8 + 0.3 \times 3.6)}$

= 3.377 ton/m³.

→ Estimation of the No. of shovel :-

Req. material handle per year = 7.8×10^6 ton

Avg. Density (δ_{avg}) = 3.377 ton/m³

Effective production per shovel per year = $B_e \times B_f \times \delta_f \times 3600 \times 7200 \times \eta \times \delta_{avg}$

No. of working shovels required $= \dfrac{7.8 \times 10^6}{5228568.576}$

$= 1.491$

$\therefore$ Actual No. of working shovels $\}= 2$

→ Estimation of the No. of Dumper :-

Effective dumper capacity = 50 ton

Ton per pass of shovel Bucket

$= B_c \times B_f \times S_f \times f_{avg}$

$= 5 \times 0.8 \times \dfrac{100}{100+25} \times 3.377$

$= 10.807$

$\therefore$ No. of passes required to load dumper $\} = \dfrac{50}{10.807} = 4.62$

$\therefore$ Actual No. of passes = 4

Total Dumper cycle time :-

$= 0.3 + 2 + \dfrac{4 \times 30}{60} + \dfrac{1}{20} \times 60 + 1.5 + \dfrac{1}{30} \times 60$

$= 0.2 + 2 + 2 + 3 + 1.5 + 2$

$= 8.8$ min

$\therefore$ No. of Dumper $= \dfrac{8.8}{0.3+2} = \underline{\underline{3.82}}$

Actual No. of Dumpers required per Shovel } = 4

Total No. of Dumpers Required } = No. of Shovel × 4

= 2 × 4 = 8

- Statement of HEMM

Sl. no.	Item	Capacity	Working	Spare	Total
1.	Shovel	$5\ m^3$	2	–	2
2.	Dumper	80 ton	8	4	12
3.	Drills	225 mm	2	–	2
4.	Dozer	350 HP	5	1	6
5.	FEL	$3\ m^3$	3	1	4
6.	Motor grader		2	1	3
7.	Crane		2	1	3
8.	Water Sprinkler		2	1	3